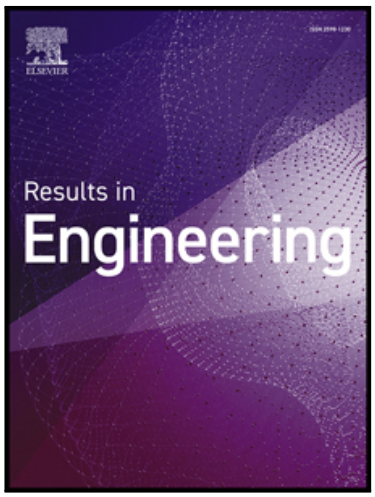
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Attention Driven Hybrid Deep Representation Learning Model for Intrusion Detection in Software Defined Networks Environment

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Highlights

- Proposes DRL-ADSDN for accurate DDoS detection in SDN
- Uses MIC–mRMR for optimal feature selection
- CRAN model captures spatial–temporal attack patterns
- RAdam optimizer improves convergence and generalization
- Includes robust preprocessing and data normalization
- Outperforms existing methods across key metrics
- Enables scalable and intelligent SDN security solution

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Attention Driven Hybrid Deep Representation Learning Model for Intrusion Detection in Software Defined Networks Environment

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Abstract

Software-Defined Network (SDN) is an innovative network architecture that enables a more centralised, programmable approach to infrastructure control, giving operators greater flexibility and programmability. In SDN, the controller, also known as the network operating system, manages network devices, executes network applications, and delivers essential network services and functions. Even with its advanced features, implementing several architectural components of SDN introduces numerous security risks and potential attack vectors. Distributed Denial-of-Service (DDoS) attacks are rapidly escalating and pose a significant risk to the Internet. Due to SDN vulnerabilities, considerable recent research has leveraged deep learning (DL) to recognise and classify DDoS attacks. DL can automatically find correlations within the data; hence, it is an efficient technique for the next generation of attack detection. This paper presents a Deep Representation Learning for Attack Detection in Software-Defined Networks (DRL-ADSDN) methodology. The aim is to achieve high-accuracy intrusion detection and classification. Initially, pre-processing includes cleaning, outlier normalisation, and data splitting, to ensure consistent and stable model training. Following this, the maximum information coefficient and minimum redundancy maximum relevance (MIC-mRMR)-based feature selection strategy is designed to identify highly informative attributes while removing the redundant ones. For intrusion detection and classification, convolutional recurrent attention network (CRAN) model is employed. The model is further optimised with the rectified Adam (RAdam) optimiser, resulting in faster convergence and improved generalisation. A comprehensive experimental analysis is conducted, and the results are examined using numerous metrics. The comparison study

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demonstrated the enhanced performance of the DRL-ADSDN technique compared to existing approaches.

Keywords: Software-Defined Networks; Threats; Feature Selection; Hybrid Classification; Adam Optimization; Deep Learning

1. Introduction

SDNs introduce a novel approach to the telecommunications network model, improving network functionality and service control through virtualisation [1]. This system allows users to configure and manage the network centrally, supporting adjustments to the larger devices connected to the Internet and optimising network traffic, which requires greater utilisation of available network resources [2]. Although this centralised control method also presents an individual point of failure: the controller. Hence, the accountability for coordinating each switch, its performance, and availability is essential for network management [3]. A DDoS attack targeting the controller itself or the network can utilise its computational and communication resources, thereby affecting the overall structure's accessibility. Earlier detection of anomalous traffic is considered superior [4]. Conventional techniques derived from static principles or signatures are limited by the constant growth in attacks, which has facilitated the rise of methods based on machine learning (ML). By evaluating statistical parameters of data flows, the methods above can recognise trends that differentiate normal activity from malicious traffic and adapt to variations in attack strategy and intensity [5]. Hence, the SDN control planes provide a real-time reduction system that enables responding to and predicting DDoS threats before controller failures, thereby enhancing network stability. Currently, DL is being developed as an effective tool for managing the massive amounts of data generated by SDN. Its ability to automatically extract complex data representations makes it a standard option for managing the intricacies of SDN data [6]. The DL method also supports deep linking. Unlike conventional ML approaches that rely on DL techniques, manual feature engineering can evaluate data at multiple levels of abstraction, uncovering hidden patterns and correlations. These methods utilise various layers of nonlinear processing units to extract both specialised and generative features, employ signal-processing strategies, and draw inspiration from neural functions [7]. DL models are widely divided into unsupervised (generative), supervised (discriminative), and hybrid techniques. Discriminative techniques are extensively exploited for tasks such as prediction and classification [8]. CNNs minimise the complexity of conventional neural networks (NNs) by using sparse interactions, parameter sharing, and supporting equivariance to transformations. These methods improve the model's performance; hence, they may introduce complexity in scalability and training. Collectively, these methods

form a robust framework for proficient feature engineering, data handling, and classification, enabling the proposed system to achieve excellent performance in detecting SDN-based intrusions [9]. Fig. 1 represents the general diagram of SDN.

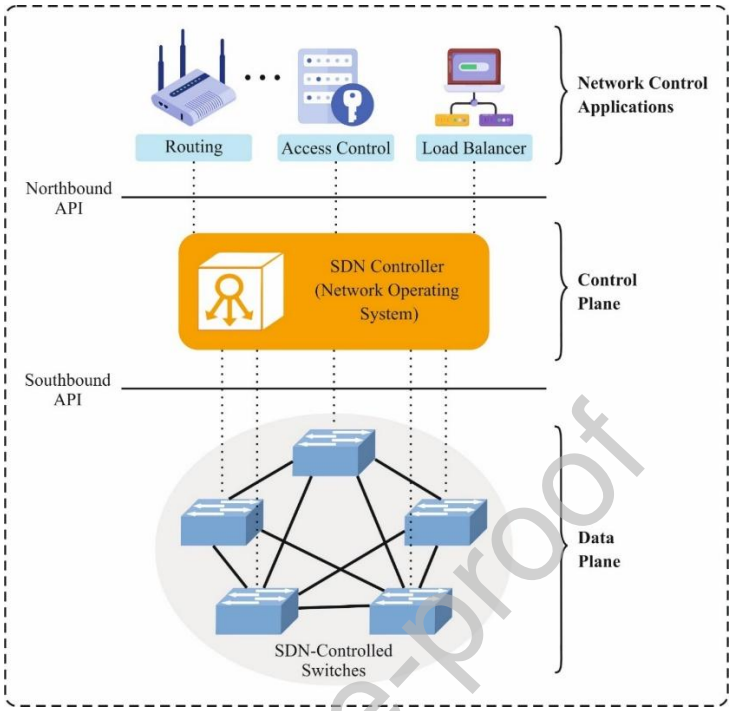


Fig. 1. General diagram of SDN

To address the challenges of evolving attack patterns in SDN, this study presents a Deep Representation Learning for Attack Detection in Software-Defined Networks (DRL-ADSDN) model. The proposed method integrates robust pre-processing via noise removal, outlier normalisation, and data splitting, optimal feature selection, and DL-based attack classification to ensure better attack detection performance. The DRL-ADSDN model is assessed on a benchmark dataset under different evaluation metrics.

Although the deep learning-based intrusion detection systems (IDSs) have achieved great success, they still suffer from some drawbacks in SDN applications. Current methods can work well on a certain benchmark dataset, but can be impacted by multiple and correlated traffic attributes, varied flow patterns, shifting attack types and variations in traffic across datasets [11]. Moreover, most of the current models focus on the classification accuracy without taking into account the cost of feature processing and deep representation learning. Although hybrid CNN, recurrent and attention-based models are able to model complementary traffic properties, the combination of feature redundancy reduction and spatial-temporal representation learning in a single model is not sufficiently explored [22]. In the case of attacks with traffic characteristics that vary significantly from the more prevalent attack classes, these limitations can be quite relevant, as demonstrated in the ToN-IoT evaluation results. Thus, a successful

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SDN intrusion detection system should minimize unnecessary traffic attributes, consider both local and temporal dependency, focus on attack-relevant representations and have reasonable computational behavior.

The novelty of DRL-ADSDN lies not at the level of introducing each learning component as a new algorithm, but at the level of integration into a framework. Before deep representation learning, the informative attributes are identified by using MIC-mRMR. The CRAN architecture subsequently combines CNN-based local feature extraction, BiGRU-based bidirectional temporal modelling and additive attention for emphasizing attack-relevant representations. RAdam is then used to tune the integrated representation-learning architecture. The framework that emerges as a result of this is thus a spatio-temporal sequence of feature reduction, spatial representation learning, modeling temporal dependencies, attention-based feature weighting and optimization. The integration that has been designed in this way is meant to compensate for complementary characteristics of network traffic and not to depend on one feature-selection or learning mechanism.

SDN: Network traffic is dynamically managed using programmable control and forwarding. This induced flow-level traffic is heterogeneous and can have redundant, correlated and temporal-dependent attributes. The goal of the proposed framework is to develop a discriminative mapping between these traffic features and normal and attack traffic classes without losing the excessive features information and without increasing the computation cost. The key contributions of this research are as follows:

- Present a feature reduction mechanism combining MIC-mRMR to obtain discriminative features while eliminating irrelevant and repetitive attributes.
- Design a CRAN, which integrates the convolutional NN (CNN), a bidirectional gated recurrent unit NN (BiGRU) with an additive attention mechanism (AAM) for capturing spatial as well as temporal relations in network traffic and highlighting significant patterns associated with malicious behaviours through attention-driven representation learning.
- Integrating the RAdam helps attain model convergence stability and generalisation, and improves classification performance.
- Conduct comprehensive experimental assessments on benchmark datasets and perform comparative analysis, highlighting that the DRL-ADSDN method performed better compared to other models across several performance metrics.

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2. Literature Survey

This section presents a summary of related work on attack detection in SDN. Polat and Polat [10] developed an SDN controller-based mechanism for detecting and mitigating DoS attacks. The approach monitors packet-flow behavior and applies threshold-based detection to identify abnormal traffic. Its effectiveness was evaluated using a POX controller and Mininet environment, demonstrating the feasibility of controller-level attack mitigation. Polat et al. [11] reviewed supervised and deep learning methods for DDoS detection in SDN environments. The study compared existing learning-based approaches across different network applications and identified challenges related to detection accuracy, adaptability, and SDN security. The review also emphasizes the need for more intelligent and adaptive mechanisms for DDoS detection and mitigation.

Kachavimath and Narayan [12] proposed a robust method for DDoS attack detection using a hybrid DL approach. For using a sophisticated dimensionality reduction procedure that relies on a consensus method to identify the eight characteristics, thereby improving identification precision and effectiveness. Mehmood et al. [13] presented the implementation of both a multilayer perceptron (MLP) and a CNN-driven convolutional approach for detecting DDoS attacks. This technique employs intricate optimisation to reduce false positives or improve DDoS detection efficacy, utilising SHAP to select a feature model and enhance detection. By assisting in classifying the features most important for spotting events, this method improves processing accuracy and consistency. Gayantha et al. [14] proposed an ML-driven architecture that efficiently distinguishes between malicious and regular traffic by leveraging engineered features such as packet rates, unique source counts, and flow counts. The structure integrates a dynamic reduction mechanism that accepts legitimate blocks or redirects malicious traffic while minimising disruption to legitimate performance. Hnamte et al. [15] proposed a method for identifying DDoS using a DNN architecture grounded in DL principles. The presented technique offers an adaptable, scalable architecture that enables meticulous analysis of network traffic to discern complex patterns indicative of DDoS attacks. To validate the method's efficiency, exact assessments are conducted using authentic traffic data. Paidipati et al. [16] presented a metaheuristic with multilayer ensemble deep reinforcement learning for a DDoS recognition model in the Cloud SDN surrounding. This model utilises metaheuristics and the DL technique to detect DDoS threats at the SDN data level. Then, the predictive system aims to use African buffalo optimiser-driven feature reduction to mitigate computational difficulty and increase the detection rate.

Al-Dunainawi et al. [17] proposed a new, optimised method that efficiently integrates Mininet, a Ryu controller, and 1D-CNNs to mitigate and identify DDoS attacks in SDN environments. This method trains a 1D-CNN using labelled network traffic data to classify abnormal patterns related to DDoS attacks efficiently. Setitra et al. [18] proposed a model that integrates CNNs and MLPs to improve the efficiency of ML-driven DDoS-detection methods in SDN environments. The presented approach employs the SHAP feature-selection model and Bayesian optimisation to tune the technique's parameters.

Aslam et al. [19] presented an Adaptive ML-driven SDN-aided DDoS attack Detection and Mitigation structure. This structure extends an SDN-allowed security module for IoT systems, using an ML-based identification technique to achieve effective mitigation and detection of DDoS threats. The method employs ML models in an adaptive, multiple-layer feed-forward scheme to successfully detect DDoS threats by assessing static attributes of the examined traffic. Swileh and Zhang [20] developed a Natural Language Processing (NLP)-based attack detection framework. The model also utilizes the Bidirectional Encoder Representations from Transformers base uncased model (BERT base uncased) with Random Forest (RF) based feature selection. Gadallah, Ibrahim, and Omar [21] presented an Autoencoder with Bidirectional Gated Recurrent Unit (AE-BGRU) based detection framework to identify and reduce DDoS attacks. Wang et al. [22] proposed an Attention Residual Stacked Autoencoder with Gated Recurrent Unit (ARSAE-QGRU) methodology by integrating attention mechanisms (AMs), residual connections, and temporal analysis to accurately detect DDoS attacks. Abid et al. [23] developed a multilevel deep neural network (DNN) integrating Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) models. Veeresh, Jaitly, and Reddy [24] presented a Trident Autoformer Fusion Network (Trident-AFNet) model integrated with LyrebirdNizar Integrated Tuner for Swish Function Optimization (LyNi-Tuner) to accurately detect and classify attacks in SDN. Chattopadhyay, Sahoo, and Jasola [25] introduced an ensemble learning (EL)-based detection framework by integrating diverse techniques to precisely detect and classify DDoS attacks. Zaidoun and Lachiri [26] developed a hybrid DNN-LSTM model enhanced with AM and Transformer methods. Palaniappan, Duraipandi, and Balasubramanian [27] proposed a dynamic behavioral profiling and anomaly detection framework to identify and mitigate security threats in real time. Leena, Karamadi, and Manjesh [28] introduced a Correlation Lotus Effect Hierarchical Dual Graph Neural Network (CLEHDGNN) approach by incorporating correlation analysis, Density-Based Spatial Clustering (DBSC), and Particle-Guided Metaheuristic Algorithm (PGMA) to accurately detect and classify cyber–physical attacks. Rbah et al. [29] presented a

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hybrid DL method by integrating LSTM with an attention layer within a Software-Defined Network (SDN) to accurately detect intrusions.

The limitations include reliance on particular dataset that may mitigate generalizability across diverse SDN scenarios, and high computational complexity in models like ARSAE-QGRU, Trident-AFNet, and CLEHDGNN, restricting real-time deployment. Extensive preprocessing is required by NLP- and BERT-based models and also various approaches concentrate mostly on DDoS detection than other types of attacks. Furthermore, it becomes difficult to comprehend decision rationale due to few methods such as AE-BGRU and hybrid DNN-LSTM. There is also a lack of standardized benchmarking across SDN datasets, affecting the comparability of results. The research gap while addressing it involves developing models that are scalable, interpretable, efficient, and capable of detecting both known and unseen attacks across heterogeneous SDN traffic patterns.

3. Proposed Model

The DRL-ADSDN model introduces an effective method for recognising and classifying attacks in SDN settings. The DRL-ADSDN method follows a systematic approach that integrates comprehensive pre-processing, dimensionality reduction, classification, and optimisation. Fig. 2 showcases the end-to-end architecture of the DRL-ADSDN method.

As shown in Fig. 2, the proposed method proceeds through a sequence of phases.

- The input dataset undergoes pre-processing, including data cleaning to remove inconsistencies, outlier normalisation to stabilise extreme values, and systematic training and testing splits to ensure unbiased model evaluation.
- Following pre-processing, the proposed model applied the MIC-mRMR model. This hybrid approach selects the most influential traffic features while concurrently removing redundant or irrelevant attributes, thereby improving the discriminative capability of the learning model.
- The selected features are then fed into the classification technique, which is built on CRAN. This architecture integrates a CNN for spatial pattern selection, a BiGRU to model temporal relations, and an AAM to highlight critical traffic behaviours that are contributing to attacks. Overall, the CRAN enables deep hierarchical representation learning, making the model highly responsive to both short-term variations and long-term traffic dynamics.
- Furthermore, the model parameters are optimised using the RAdam optimiser to ensure stable convergence and better generalisation across various attack scenarios.

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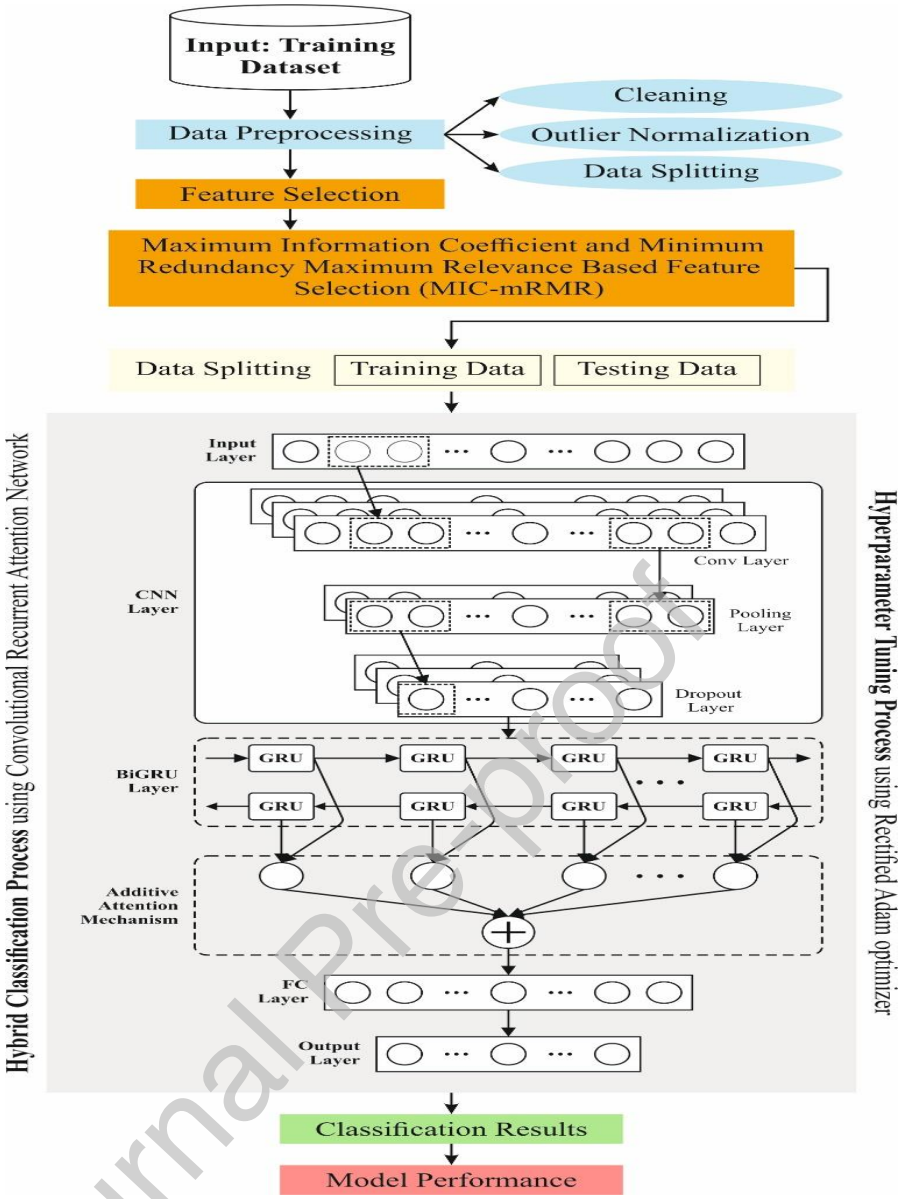


Fig. 2. End-to-end architecture of the DRL-ADSDN model

DRL-ADSDN's individual components are chosen based on the complementary properties of network-flow data. MIC-mRMR is used at the input stage because it is possible that there are correlated and redundant attributes in a network-flow dataset that could cause a heavy burden when learning the subsequent step. Local relationships between the selected traffic attributes are then identified using CNN. Malicious traffic can have temporal dependencies, and thus these are captured by using BiGRU to learn forward and backward dependencies. The additive attention mechanism also gives different weights to the learned temporal representations so that the elements related to an attack are more emphasized when classified.

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3.1. Pre-processing Pipeline

In the pre-processing step, cleaning, outlier normalisation, and data splitting are applied to ensure consistent and stable model training.

Dataset Cleaning: Data cleaning involves removing samples or null values. The sample comprises any inaccessible values that were deliberated. Additionally, data cleaning was performed to extract the lesser contributing attributes. The attributes that do not correlate with the outcome variable are eliminated.

$$f(x) \leftarrow d_i = 0, x_i \leq 0 \tag{1}$$

Outlier Normalisation Utilising the IQR approach: To compare the range of outliers against every aspect, then discover specific values that are enormously out of range compared to many samples in the feature column [30]. By checking the value range for each aspect, a box plot was used to show the quartile range for each attribute. Many attributes have specific ranges of input values across the entire dataset; here, a few of them show higher values or outliers that need to be handled appropriately. Nevertheless, outlier removal is assumed to employ the IQR model.

$$IQR = q_3 - q_1 \tag{2}$$

IQR is implemented to encompass the values of feature outliers. It is computed to arrange the data values in ascending order and to separate them into four equal proportions. The value of each attribute after covering it using the IQR approach was extended. These values comprise all values in the normalised data. Attributes that comprise outliers, but employ the IQR approach for each attribute, normalise each data point. This produces noise-free data and shares it with equivalent ranges.

Data Splitting:

T The data is split into a training, validation and testing set prior to building the model. The training subset is 70% of samples available, and used to train the model parameters. A validation subset is independent from the training subset and is used to track learning behaviour and tune the model hyperparameters, such as the optimisation settings, during model development. The testing subset is not used at all during development of the model, but is only used to evaluate the final model. No test samples are used in the parameter optimisation process. The separation makes it possible to report the result of the testing as the model's performance on unseen test samples. The validation subset is used exclusively during the process of creating the model and choosing the hyperparameters, while the test subset stays separate until the end to test the final performance of the model. The reported testing metrics are, therefore, not used to optimise the model parameters. In order to ensure a common

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evaluation protocol, the same preprocessing and feature-selection steps are used for the same partition of the data.

3.2. Optimal Feature Selection

The MIC-mRMR-based dimensionality reduction technique is used to choose beneficial features while eliminating the redundant ones [31]. This is selected for its efficiency in capturing linear and nonlinear associations between attack labels and features. This also provides higher sensitivity than correlation-based filters. Additionally, the mRMR model maximises relevance while minimising redundancy to obtain a compact yet highly discriminative feature subset. Also, the model offers greater robustness to intrinsic, nonlinear traffic patterns commonly observed in SDN environments, unlike conventional models such as chi-square, ANOVA, or ReliefF. The integration not only improves the detection accuracy but also mitigates computational overhead, making it appropriate for large-scale network intrusion datasets.

In the study of attack detection in SDN, many factors influence it. This method incorporates essential factors and data types.

The maximal information coefficient (MIC) is used to determine the nonlinear dependence among attributes. The calculation of MIC is specified:

$$I(x,y) = \sum_{x\in X}\sum_{y\in Y} p\,(x,y)\log\left(\frac{p(x,y)}{p(x)p(y)}\right) \tag{3}$$

Here, $p(x,y)$ indicates the joint probability density of variable X and Y; $I(x,y)$ implies the mutual information (MI) coefficient between X and Y; $p(x)$ and $p(y)$ indicate the marginal probability density of variable X and Y.

The feature selection approach called mRMR, which uses MI scores to select attributes. The objective is to penalise the correlation of attributes through their repetition in the existence of other chosen attributes.

$$D(S,c) = \frac{1}{|S|}\sum_{f_i\in S} I\,(f_{\mathrm{i}},c) \tag{4}$$

The feature redundancy in set S is the mean of the MI values for features f_i and f_j.

$$R(S) = \frac{1}{|S|^2}\sum_{f_i f_{j\epsilon S}} I\left(f_{\mathrm{i}},f_j\right) \tag{5}$$

To incorporate the constraints of the above dual calculations, a parameter $c\varphi$ is introduced to enhance D and R concurrently.

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$$\emptyset = \max \left[\frac{1}{|S|} \sum_{f_i \in S} I\left(f_i, c\right) - \frac{1}{|S|^2} \sum_{f_i, f_j \in S} I\left(f_i, f_j\right) \right] \tag{6}$$

An incremental search approach is used to attain near-optimal attributes.

$$\underset{x_j \in X - S_{m-1}}{max} \left[I\left(x_j, c\right) - \frac{1}{m-1} \sum_{x_i \in S_{m-1}} I(x_j, x_i) \right] \tag{7}$$

Here, X denotes the set of each feature, S_{m-1} denotes the chosen subset of features, and its feature subset comprises $m-1$ attributes. The initial attribute in the remaining feature set is assigned to the subset of features rather than the later feature; it cannot guarantee that the outcome after including the attributes. MIC is a subset of features, and the remaining attributes are sorted using the mRMR approach. To add a primary feature to the rest of the feature set within the subset of features and put them in a method.

3.3. Threat Classification Module

For attack detection and classification, the DRL-ADSDN model used a CRAN, emphasising the crucial patterns representing malicious activities. The model is chosen for its effectiveness in capturing both spatial patterns and temporal dependencies, which are essential for modelling dynamic network traffic [32]. The AAM emphasises that the most malicious-relevant features enhance interpretability and precision. BiGRU is computationally lighter than LSTM while presenting comparable performance, making it appropriate for large-scale SDN datasets. Moreover, the integrated model effectively extracts multi-level temporal–spatial representations, resulting in superior detection accuracy compared with shallow classifiers and single-stream deep models. Also, the robustness against subtle and growing attack signatures is improved by the attention-driven representation. The fusion search space included various combinational strategies, namely simple feature concatenation, learnable weighted summation, and bilinear pooling, to explore diverse and expressive ways of integrating CNN, BiGRU, and attention-based representations.

CNN: 1D CNN is a NN framework that operates to eliminate attributes from short, fixed-length inputs across the entire dataset and is frequently employed to process textual or 1D time-series data. First, convolution operations were applied to eliminate salient aspects of the sequence; then, a convolution kernel was applied to the input in a sliding-window fashion to create a novel representation [30].

$$\xi[j] = (X \cdot W)[j] = f\left(\sum_{k=0}^{K-1} X\left[j \times T_s + k\right] \cdot W[k] + b_{cnn} \right) \tag{8}$$

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Here, $\xi[j]$ indicates the j_{th} component of the outcome sequence. $X[j+k]$ implies the $j+k_{th}$ component of input order; here, j signifies the existing place of the convolution operation, and k represents the slide of the convolution size. The convolution kernel multiplies element-wise with a segment of the input order (from j to $j+K-1$) and adds the j_{th} component of the outcome series. T_s denotes time-step; $W[k]$ implies k weighted of convolution size is leveraged for element-by-element multiplication with input order; K denotes dimension of convolution kernel; b_{cnn} signifies biased term, and f refers to activation function, which presents nonlinear aspects.

Subsequently, the mapping feature is reduced in size using pooling, which not only reduces computational overload but also eliminates essential attributes.

$$x[j] = \max\left(X[j \times T_s : (j+1) \times T_s]\right) \tag{9}$$

Here, $x[j]$ signifies the primary j component of the pooled outcome series, the order of the feature attained by the pooling operation. $X[j \times T_s : (j+1) \times T_s]$ denotes a sub-sequence chosen in input order; the starting index of sub-sequence $j \times T_s$ and end index $(j+1) \times T_s$. This creates an indexing operation that is frequently leveraged to intercept a specific portion of a sequence. Fig. 3 demonstrates the infrastructure of the CNN technique.

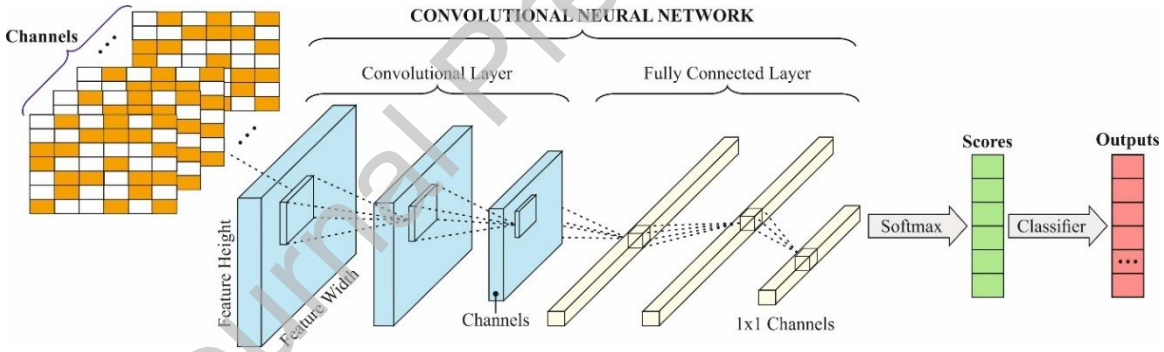


Fig. 3. Structure of the CNN technique

BiGRU: GRU and LSTM both maintain substantial historical context through gating mechanisms. Here, the combination of forget and input gates in an LSTM is compressed into an upgrade gate, which contains the upgrade gate z and the resetting gate r. GRU have no kernel state and calculates outcomes instantly. Here, h_{t-1} implies the preceding moment state; χ_i refers to the existing moment input of the GRU module; r_t signifies the resetting gate, and is capable of leveraging the candidate state $\tilde{h}_t$ on h_{t-1}; z_t indicates the upgrade gate, and h_t implies the candidate value of the output after the resetting gate process.

$$r_t = sigmoid(U_r h_{t-1} + W_r x_t + b_r) \tag{10}$$

$$z_t = sigmoid(U_z h_{t-1} + W_z x_t + b_z) \tag{11}$$

$$\widetilde{h_t} = \tanh\left(U_c(r_t h_{t-1}) + W_c x_t + b_c\right) \tag{12}$$

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$$h_t = z_t h_{t-1} + (1 - z_t)\tilde{h}_t \quad (13)$$

Now, U_r and W_r implies weighted matrices of outcome of $t-1$ moment and inputting t moment to resetting gate, correspondingly, and b_r implies bias of resetting gate; U_z and W_z refers to weighted matrices of outcome of $t-1$ moment and the input t moment to upgrade gate, correspondingly, and b_c refers to biased candidate hidden state, and U_c and W_c refers to weighted matrices of outcome of $t-1$ moment and the input t moment to candidate concealed state, v_a signifies learned weight vector.

The sigmoid function regulates the series of values for reset and upgrade gates with the expression $(x) = 1/(1+e^{-x})$. The resetting gate determines the integration of χ_i and h_{t-1} and inclines toward 0 as the ratio of h_{t-1} decreases. The upgrade gate z_t proves the dimension of h_{t-1} into the existing moment.

Bi-GRU separates the outputs of two GRU layers into forward and backwards directions.

$$\overrightarrow{h_t} = GRU_f\left(x_t, \vec{h}_{t-1}\right) \quad (14)$$

$$\overleftarrow{h_t} = GRU_b\left(x_t, \vec{h}_{t-1}\right) \quad (15)$$

$$h_t = \vec{h}_t \oplus \overleftarrow{h_t} \quad (16)$$

Now, GRU_f and GRU_b imply incorporation of forward and reverse GRU functions, respectively; $\oplus$ they refer to the vector splice operation, and $\vec{h}$ and $\vec{h}_t$ imply forward and backward GRU passes, respectively. Fig. 4 depicts the architecture of the CRAN approach.

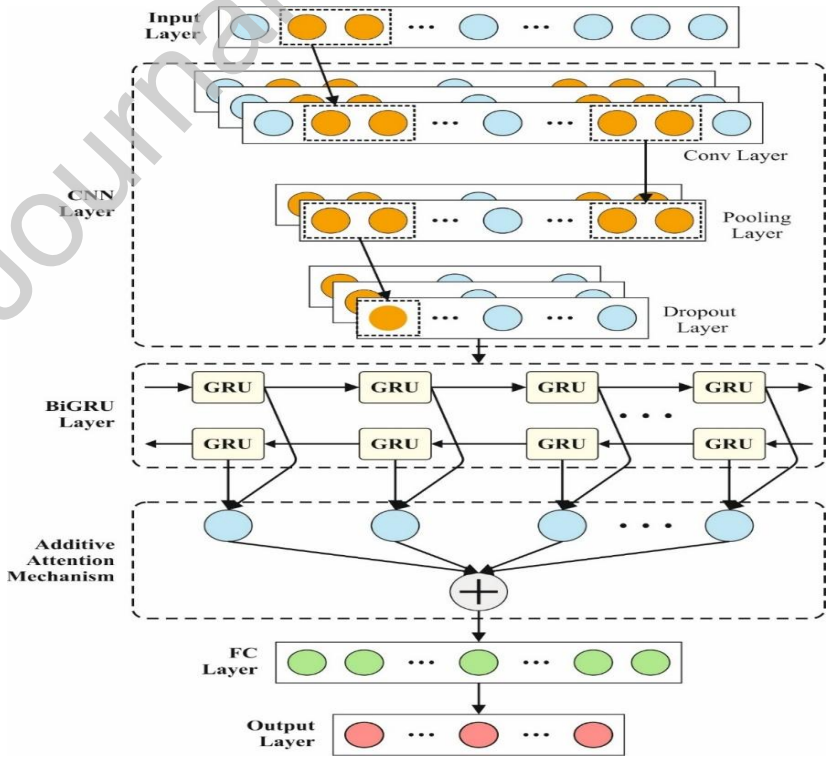


Fig. 4. Architecture of the CRAN approach

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AAM: An AM that primarily employs additive attention. It allows the method to flexibly assign varying weights to diverse segments in the input order, enabling vital components in data processing and the activation of altered attributes via a sigmoid function to portray the similarity relations between dual attributes, effectively managing nonlinear relationships.

$$y_i = DecoderOutput(s_t, c_t) \tag{17}$$

Now, y_i represents the outcome of the additive attention decoder, s_t implies the concealed vector of additive attention decoding at time-step t, y_i refers to the result of the decoder at time-step i, and c_t is attained by decoding at time-step t, which depends upon the outcome of encoded $(h_0, h_1, \dots, h_t)$, which represents the weighted addition of encoder outcomes.

$$c_t = \sum_{\mathrm{i}=1}^{\mathrm{T}} a_{t,i}\, h_i \tag{18}$$

$$a_{t,i} = \frac{\exp\,(e_{t,i})}{\Sigma_{j=1}^{T} \exp\,(e_{t,j})} \tag{19}$$

$$e_{t,i} = v_a^T \tanh\,(W_a h_i + U_a s_{t-1}) \tag{20}$$

Here, $e_{t,i}$ implies attention score from decoding at t_{th} time-step to encoded at i_{th} time-step; the decoder outcome at existing position relies on the hidden state of decoding s_{t-1}, the encoder outcome of encoding at i_{th} time-step h_i; W_a and U_a refer to learned parameters, and v_a signifies learned weight vector. Algorithm 1 depicts the CRAN method. Table 1 describes the hyperparameters of the CRAN model.

Algorithm 1: CRAN model

Input: Network traffic data

Output: Attack or benign classification

Step 1: Preprocessing

Segment sequences and normalize traffic features for model input.

Step 2: CNN Feature Extraction

Capture local spatial patterns by applying 1D convolution and utilize pooling for reducing dimensionality and retaining salient features.

Step 3: BiGRU Temporal Modeling

Send CNN features to BiGRU layers for learning forward and backward temporal dependencies.

Step 4: AAN

The most malicious-relevant features are highlighted by computing the attention weights and the context vectors are generated by representing crucial temporal–spatial relations.

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Step 5: Feature Fusion Integrate CNN, BiGRU, and attention outputs. **Step 6:** Classification Layer Send fused features to the dense layer for predicting attack class labels. **Step 7:** Output Return predicted class indicating benign traffic or specific attack type.

Table 1 Hyperparameters of the CRAN method

Component	Hyperparameter	Value
1D CNN	CONV_LAYER_NO	2
	KERNEL_SIZES	3, 5
	FILTER_NUM	64, 128
	ACTIVATION	ReLU
	POOL_LAYER	MAX_POOL (size=2)
BiGRU	LAYER_NO	1-2
	HIDDEN_UNITS	128
	DROPOUT	0.3
Additive Attention	ATTEN_DIMEN	128
	ACTIVATION	Tanh
Training	OPTIMIZER	Radam
	LEAR_RATE	0.001
	BATCH_SIZE	64
	EPOCHS	50

4. Dataset Specification and Measurement Metrics

In this section, the dataset description and the measures used to evaluate the model performance are presented.

4.1. Dataset Summary

The experimental experimental sizes of data sets are summarized as follows. As for the number of samples, the InSDN dataset has 10,500 network-flow records, the ToN-IoT dataset has 119,957 samples and the CICDDoS-2019 has 9,000 samples. The SDN-specific benchmark used is InSDN, and ToN-IoT and CICDDoS-2019 are complementary for the network and attack-traffic distributions. The reported execution times are therefore to be evaluated in conjunction with the characteristics and experimental setup of the data sets.

The performance of the DRL-ADSDN model is evaluated using three datasets: the InSDN dataset [33], the ToN-IoT dataset [34], and the CICDDoS-2019 dataset [35]. InSDN is a complete SDN dataset for intrusion detection system assessment. The novel dataset includes benign and several attack classes that may occur across various aspects of the SDN standard. In SDN, 10500 network flow records contain multiple attacks involving exploitation, brute-

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force, probing, DoS, web applications, DDoS, and botnets. Moreover, the normal traffic on the created information covers diverse popular application services such as DNS, HTTP, FTP, HTTPS, SSL, Email, SSH, etc. The dataset is generated using four virtual machines (VMs). This dataset is widely used to develop and benchmark DL and ML techniques for detecting anomalies and cyberattacks in modern SDN environments. The total number of 79, but 10 features are chosen, is revealed in Fig. 5. This dataset is part of the ToN-IoT dataset gathered by the Cyber Range Lab of UNSW Canberra. It comprises network traffic data from simulated IIoT and IoT environments for ML and intrusion detection tasks. This file encompasses network traffic data taken from simulated IoT and IIoT devices in a smart home setting. It involves labelled flows of both malicious and benign actions, gathered employing Argus and Bro (Zeek) tools. The dataset is intended for ML, intrusion detection, and anomaly detection development. The dataset contains 119957 samples under nine categories.

SDN specific benchmark used in this study as it is the only dataset that was created for intrusion detection in an SDN environment, and contains network-flow records of multiple attack categories and benign traffic. ToN-IoT and CICDDoS-2019 are also proposed to investigate the effectiveness of the learned representation in various network and IoT traffic distribution scenarios.

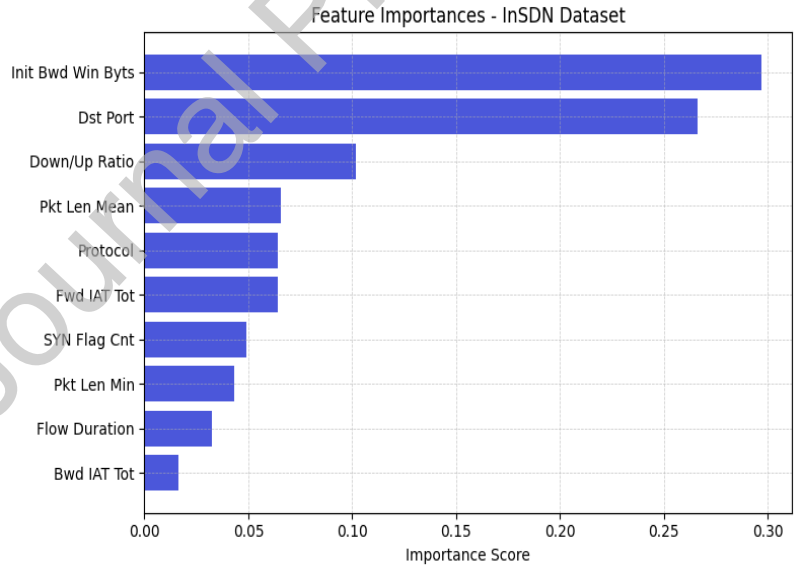


Fig. 5. Feature importance of the InSDN dataset

The total number of 63, with 10 features chosen, is shown in Fig. 6.

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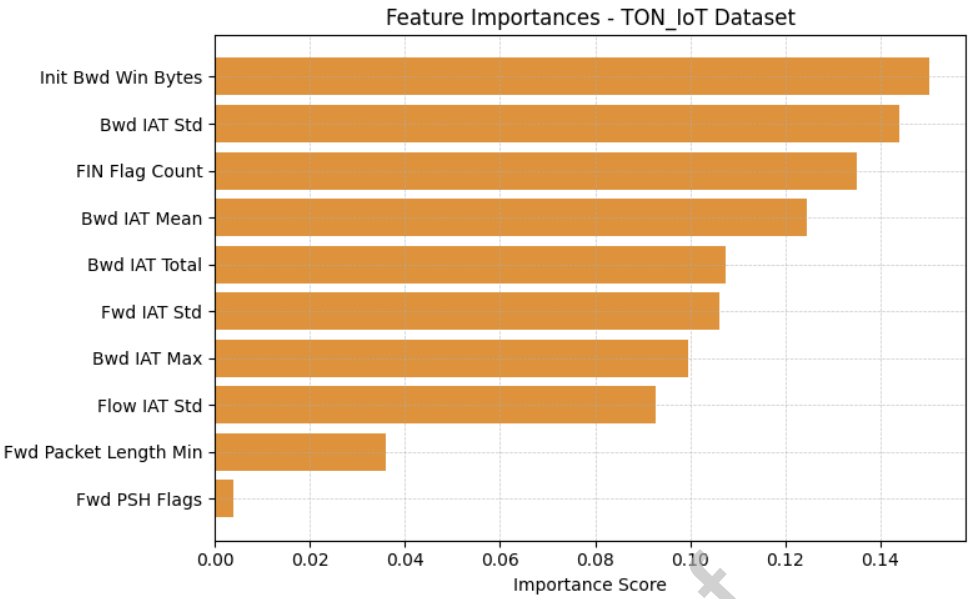


Fig. 6. Feature importance of ToN-IoT dataset

The CICDDoS-2019 dataset is a large, labelled network-traffic dataset that involves both normal traffic as well as multiple real-world cyberattacks, namely botnet, infiltration, DoS, DDoS and brute force. It provides detailed, flow-driven feature extraction from raw pcap files, making it highly suitable for training and evaluating DL- and ML-based intrusion detection systems. The dataset is widely used as a benchmark for cybersecurity research due to its realistic traffic patterns, rich feature set, and comprehensive attack coverage. The dataset contains 9000 samples under 18 classes. The total number of 77, but 10 features are chosen, is shown in Fig. 7.

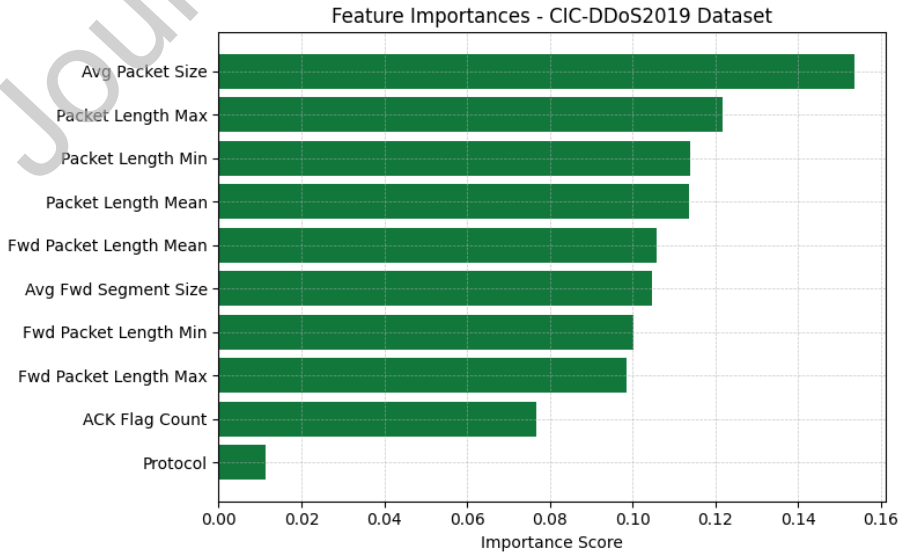


Fig. 7. Feature importance of CICDDoS-2019 dataset

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4.2. Validation Measures

The proposed approach used to calculate numerous measures, namely Accuracy, Recall, Precision, and F1 score.

Accuracy

It assesses the total number of instances against the actual instances.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{21}$$

Here, TP refers to the true positive, TN indicates the true negative, FN depicts False negative, and FP depicts false positive.

Precision

Precision is the ratio of the overall count of positive instances to the overall positive sample counts.

$$Precision = \frac{TP}{TP+FP} \tag{22}$$

Recall

Recall is the percentage of positive instances among all instances.

$$Recall = \frac{TP}{TP + FN} \tag{23}$$

F-Score: F assess the symmetrical centre between recall and precision.

$$F - Score = \frac{2*Precision*Recall}{Precision+Recall} \tag{24}$$

MCC: MCC is a metric for calculating the execution of binary classifiers.

$$MCC = \frac{(TP \cdot TN)-(FP \cdot FN)}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}} \tag{25}$$

5. Experimental Result Analysis

In this study, the performance of the DRL-ADSDN approach is evaluated across three datasets: InSDN, ToN-IoT, and CICDDoS-2019. The technique is simulated using Python 3.6.5 on a PC with an i5-8600k, 250GB SSD, GeForce 1050Ti 4GB, 16GB RAM, and 1TB HDD. Parameters include a learning rate of 0.01, ReLU activation, 50 epochs, 0.5 dropout, and a batch size of 5. Fig. 8 displays the correlation matrix for the DRL-ADSDN methodology on the SDN dataset, emphasising the strength of the relationships among major network-traffic features. A strong positive correlation is observed among Flow Duration, Fwd IAT Total, and Bwd IAT Total, suggesting their close interdependence in shaping temporal flow behaviour. Features like Protocol and Packet Length Min also reveal notable correlations, reflecting their shared influence on traffic patterns. Meanwhile, SYN Flag Count and Down/Up Ratio show moderate

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negative correlations with various features, suggesting their importance in distinguishing abnormal traffic.

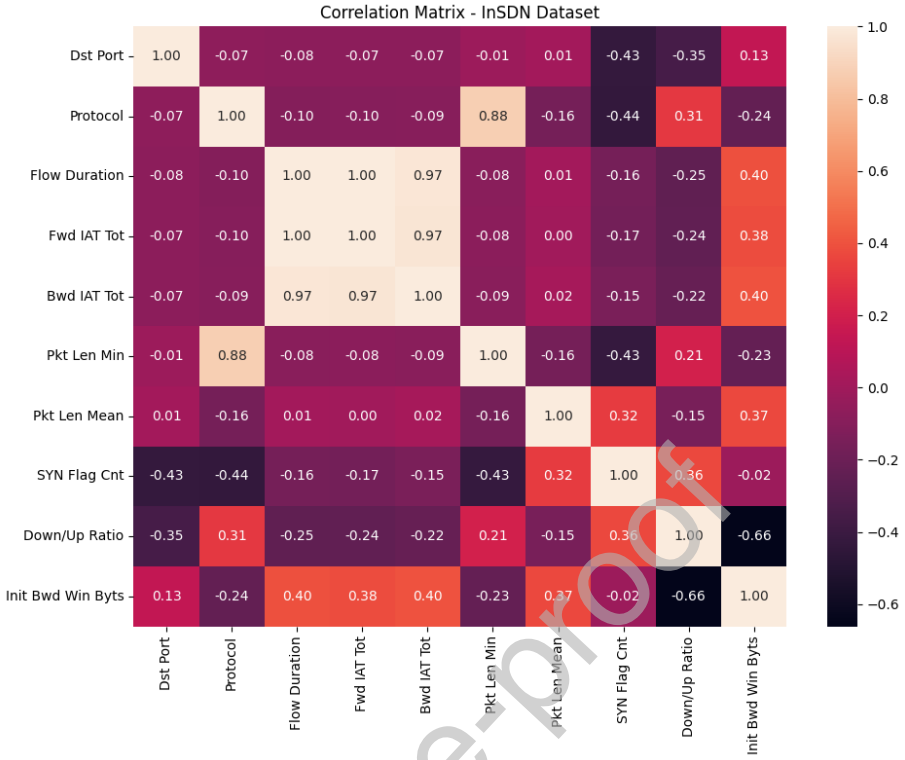


Fig. 8. Correlation matrix of InSDN dataset

Fig. 9 shows that the correlation matrix of the DRL-ADSDN methodology with the ToN-IoT dataset exhibits strong positive relations among most inter-arrival time (IAT) features, namely Fwd IAT Std, Bwd IAT Mean, and Bwd IAT Max, indicating their collective influence on flow timing behaviour. Fwd PSH Flags also correlate significantly with various IAT-driven metrics, suggesting their significance in distinguishing traffic flow features.

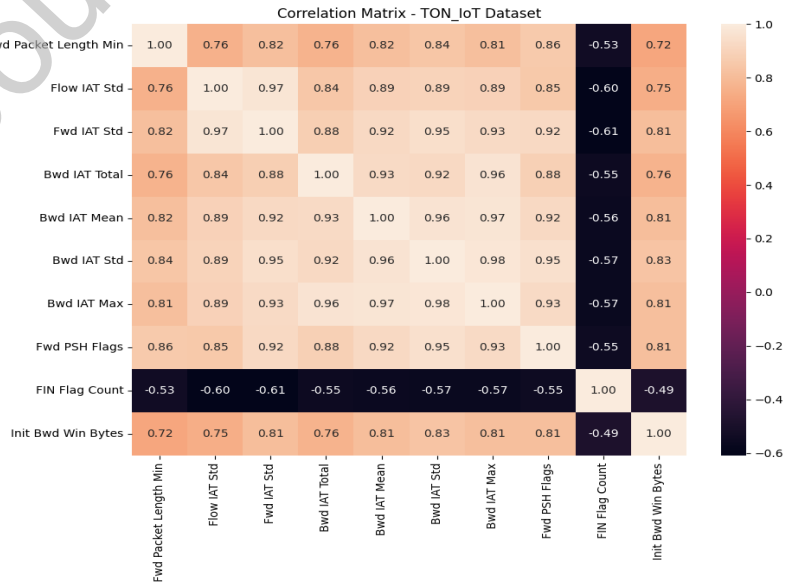


Fig. 9. Correlation matrix of ToN-IoT dataset

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In contrast, FIN Flag Count shows consistent negative correlations with most characteristics, underscoring its potential as a discriminative indicator of malicious or abnormal behaviour. Overall, the matrix provides insights into feature redundancy and helps guide efficient feature selection for IoT-driven intrusion detection techniques.

Fig. 10 depicts the correlation matrix of the DRL-ADSDN method on the CICDDoS-2019 dataset, revealing very high positive correlations among packet–length–based features, indicating robust redundancy within flow-size features.



Fig. 10. Correlation matrix of CICDDoS-2019 dataset

The attack-detection results of the DRL-ADSDN method on the InSDN dataset are clearly shown in Table 2. The experimental results indicate that the DRL-ADSDN model performs efficiently in recognition. With 70%TRAP, the DRL-ADSDN methods reach an average $accur_y$ of 99.11%, $preci_n$ of 97.27%, rec_l of 97.30%, F_{score} of 97.27%, and AUC_{score} of 98.38%. In addition, with 30%TESP, the DRL-ADSDN model gives an average $accur_y$ of 99.31%, $preci_n$ of 97.88%, rec_l of 97.95%, F_{score} of 97.91%, and AUC_{score} of 98.77%.

Table 2 Attack detection of the DRL-ADSDN approach under the InSDN dataset

Class Labels	$\boldsymbol{Accur_y}$	$\boldsymbol{Preci_n}$	$\boldsymbol{Rec_l}$	$\boldsymbol{F_{Score}}$	$\boldsymbol{AUC_{Score}}$
TRAP (70%)					
Normal	99.13	96.97	98.60	97.78	98.93
DDoS	99.37	98.07	98.64	98.35	99.09
Probe	98.84	98.58	95.23	96.87	97.45
DoS	99.24	97.76	96.82	97.29	98.23
Brute force attack	98.68	93.84	97.01	95.40	97.98
Web attack	99.40	98.40	97.48	97.94	98.61
Average	**99.11**	**97.27**	**97.30**	**97.27**	**98.38**
TESP (30%)					

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Normal	99.27	96.77	99.30	98.02	99.28
DDoS	99.43	98.84	98.19	98.51	98.96
Probe	99.14	99.00	96.60	97.79	98.18
DoS	99.40	98.68	97.19	97.93	98.48
Brute force attack	99.30	97.02	98.28	97.64	98.88
Web attack	99.33	96.99	98.13	97.56	98.82
Average	**99.31**	**97.88**	**97.95**	**97.91**	**98.77**

Fig. 11 presents the comparative error rates of the DRL-ADSDN model on the InSDN dataset. The outcomes indicate that the LSTM and CNN techniques have yielded worse error rates of 10.20 and 9.77, respectively. In the meantime, the BiLSTM, AetherNet-ID, and DNN methods have achieved slightly lower error rates of 5.42, 4.71, and 3.57, respectively. Moreover, the GRU, CNN-LSTM, and Transformer models have achieved reasonable performance, with error rates of 3.55, 2.41, and 1.87, respectively. However, the DRL-ADSDN model demonstrates strong performance, with an error rate of 0.69.

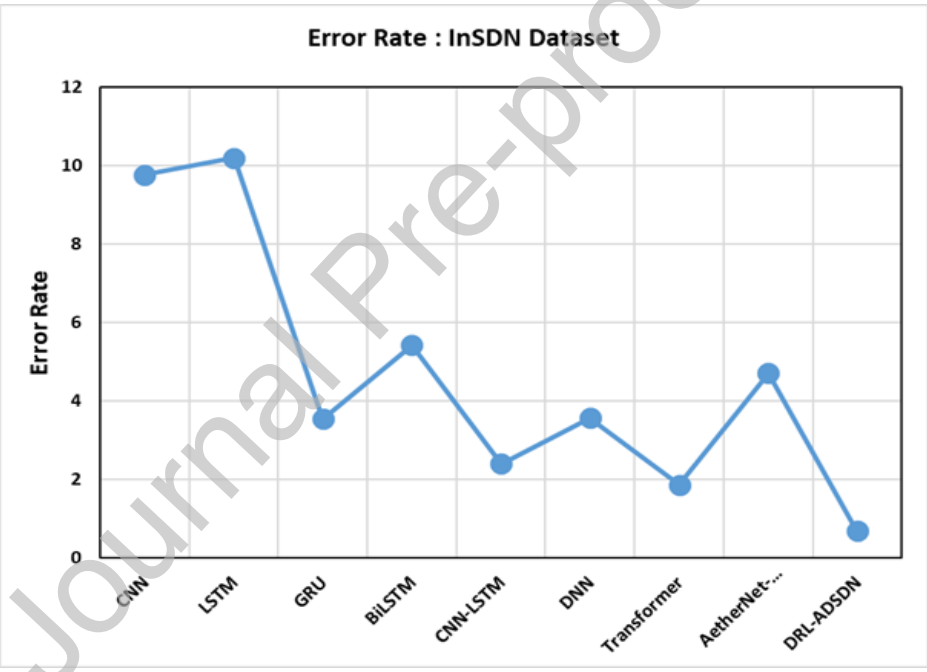


Fig. 11. Error rate of DRL-ADSDN approach under InSDN dataset

Table 3 and Figs. 12-13 indicate the comparison analysis and execution time (EXT) results of the DRL-ADSDN model on the InSDN dataset are compared with other techniques. The results indicate that the DRL-ADSDN method achieves an $accur_y$, $preci_n$, rec_l, F_{score}, and EXT of 99.31%, 97.88%, 97.95%, 97.91%, and 2.67s, which is lower than those of different approaches. On the other hand, the CNN, LSTM, GRU, BiLSTM, CNN-LSTM, DNN, Transformer, and AetherNet-ID methodologies attained increased values.

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The main challenges of DRL-ADSDN lie in evaluating the MIC features using mRMR and CNN-BiGRU-attention representation-learning architecture. In the deep learning stage, the traffic is processed dimensionality-wise by CRAN, and the dimensionality will be reduced by MIC-mRMR before the processing. At the inference stage, the main operations are the convolutional transformations, recurrent state updates and calculation of attention weights. Thus, the feature reduction is expected to decrease the unnecessary processing and the hybrid architecture is expected to have a richer spatial-temporal representation. The present study measures and compares computational behaviour in terms of execution time, not the CPU usage of the controller or energy consumption. A resource profiling at the controller level is therefore considered to be a promising avenue for further deployment experiments.

Table 3 Comparison analysis of DRL-ADSDN approach under InSDN dataset

Model	$Accur_y$	$Preci_n$	Rec_l	F_{Score}	**EXT (sec)**
CNN	90.23	94.56	96.54	93.82	7.40
LSTM	89.80	93.88	89.31	89.01	4.77
GRU	96.45	96.67	94.42	91.42	3.49
BiLSTM	94.58	89.46	91.30	93.15	7.46
CNN-LSTM	97.59	90.88	91.28	92.41	7.52
DNN	96.43	92.20	90.54	96.42	5.58
Transformer	98.13	89.75	90.37	92.85	4.72
AetherNet-ID	95.29	89.74	90.13	94.31	3.36
DRL-ADSDN	99.31	97.88	97.95	97.91	2.67

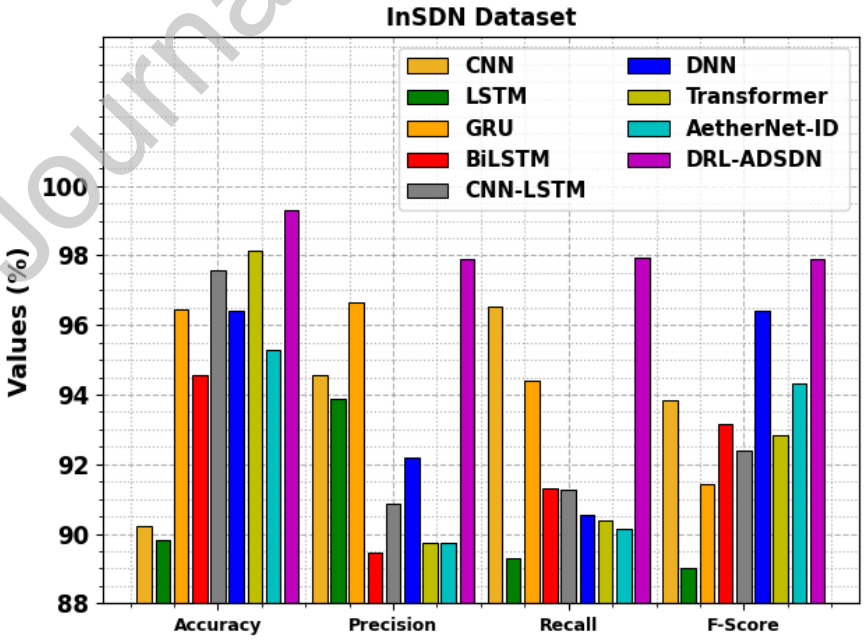


Fig. 12. Comparison analysis of DRL-ADSDN approach under InSDN dataset

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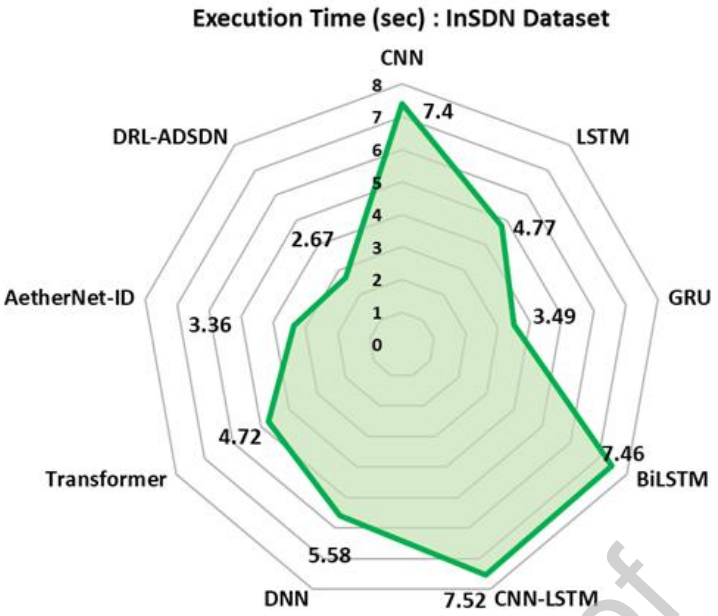


Fig. 13. EXT outcome of DRL-ADSDN approach under InSDN dataset

Table 4 depicts the ablation study analysis of the DRL-ADSDN methodology under the InSDN dataset. The CRAN+MIC-mRMR model attained an $accur_y$ of 98.66%, $preci_n$ of 97.18%, rec_l of 97.28%, and F_{score} of 97.11%, serving as a baseline to evaluate the impact of optimization. When Adam hyperparameter tuning was applied in the DRL-ADSDN model with CRAN and MIC-mRMR feature selection, performance improved to an $accur_y$ of 99.31%, $preci_n$ of 97.88%, rec_l of 97.95%, and F_{score} of 97.91%, illustrating that fine-tuning the optimizer significantly improves the predictive capability and overall classification efficiency of the model.

Table 4 Ablation study evaluation of the DRL-ADSDN methodology under InSDN dataset

Model	$\boldsymbol{Accur_y}$	$\boldsymbol{Preci_n}$	$\boldsymbol{Rec_l}$	$\boldsymbol{F_{Score}}$
CRAN+MIC-mRMR (With feature selection without Adam Hyper parameter tuning)	98.66	97.18	97.28	97.11
DRL-ADSDN (CRAN with MIC-mRMR feature selection and Adam Hyper parameter tuning)	99.31	97.88	97.95	97.91

The attack-detection outcomes of the DRL-ADSDN method on the ToN-IoT dataset are presented in Table 5. The investigational values suggest that the DRL-ADSDN approach can recognise procedures. With 70%TRAP, the DRL-ADSDN approach attains an average $accur_y$ of 99.47%, $preci_n$ of 93.60%, rec_l of 87.88%, F_{score} of 89.24%, and AUC_{score} of 93.73%. Additionally, with 30%TESP, the DRL-ADSDN approach attains an average $accur_y$ of 99.49%, $preci_n$ of 94.42%, rec_l of 89.09%, F_{score} of 90.59%, and AUC_{score} of 94.34%.

Table 5 Attack detection of DRL-ADSDN approach under ToN-IoT dataset

Class Labels	$Accur_y$	$Preci_n$	Rec_l	F_{Score}	AUC_{Score}
TRAP (70%)					
Normal	98.69	98.87	99.13	99.00	98.49
MiTM	99.76	76.32	24.27	36.83	62.12
DoS	99.57	95.70	94.85	95.27	97.32
DDoS	99.62	96.01	96.22	96.11	98.00
Password	99.54	94.80	96.00	95.40	97.86
Injection	99.54	94.83	95.87	95.34	97.80
XSS	99.57	95.75	95.52	95.64	97.65
Ransomware	99.49	95.27	94.35	94.81	97.05
Backdoor	99.47	94.84	94.71	94.78	97.22
Average	**99.47**	**93.60**	**87.88**	**89.24**	**93.73**
TESP (30%)					
Normal	98.68	98.90	99.08	98.99	98.50
MiTM	99.80	82.05	32.99	47.06	66.49
DoS	99.60	95.20	95.85	95.53	97.81
DDoS	99.59	96.12	95.86	95.99	97.83
Password	99.61	95.39	96.86	96.12	98.31
Injection	99.54	95.68	94.88	95.28	97.33
XSS	99.61	95.95	96.32	96.13	98.05
Ransomware	99.49	95.14	94.83	94.98	97.28
Backdoor	99.54	95.35	95.12	95.24	97.45
Average	**99.49**	**94.42**	**89.09**	**90.59**	**94.34**

Analysis of the ToN-IoT results by class shows a significantly weaker detection performance in the MiTM class than in other attack classes. The trained representation fails to separate this attack category as well as the rest of the classes, with the MiTM recall being 24.27% for training and 32.99% for testing. This behaviour implies that the flow-level properties of MiTM traffic can be similar to those of other traffic classes and that discriminating after a feature reduction is even more challenging. The result also shows that a high accuracy does not necessarily mean a high accuracy for each attack category.

Fig. 14 shows the comparative error rates of the DRL-ADSDN approach on the ToN-IoT dataset [34, 35]. The outcomes demonstrated that the AetherNet-ID, CNN, and DNN models exhibited the worst error rates of 8.35, 6.96, and 6.49, respectively. Simultaneously, the BiLSTM and Transformer techniques have achieved slightly lower error rates of 5.13 and 4.68, respectively. Additionally, the GRU, LSTM, and CNN-LSTM models achieve considerable results with error rates of 4.59, 3.00, and 2.56, respectively. Likewise, the DRL-ADSDN approach yields promising results, with the lowest error rate of 0.51.

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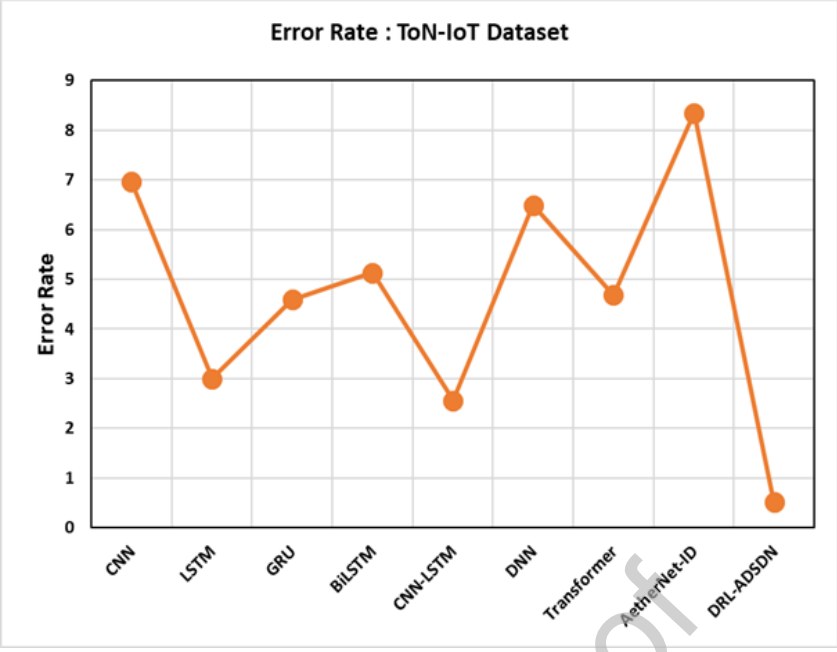


Fig. 14. Error rate of DRL-ADSDN approach under ToN-IoT dataset

Table 6 and Figs. 15-16 demonstrate the comparison and EXT assessment of the DRL-ADSDN approach under the ToN-IoT dataset. The results demonstrate that the DRL-ADSDN method achieves an $accur_y$, $preci_n$, rec_l, F_{score}, and EXT of 99.49%, 94.42%, 89.09%, 90.59%, and 1.09, while the CNN, LSTM, GRU, BiLSTM, CNN-LSTM, DNN, Transformer, and AetherNet-ID methodologies attained enhanced values.

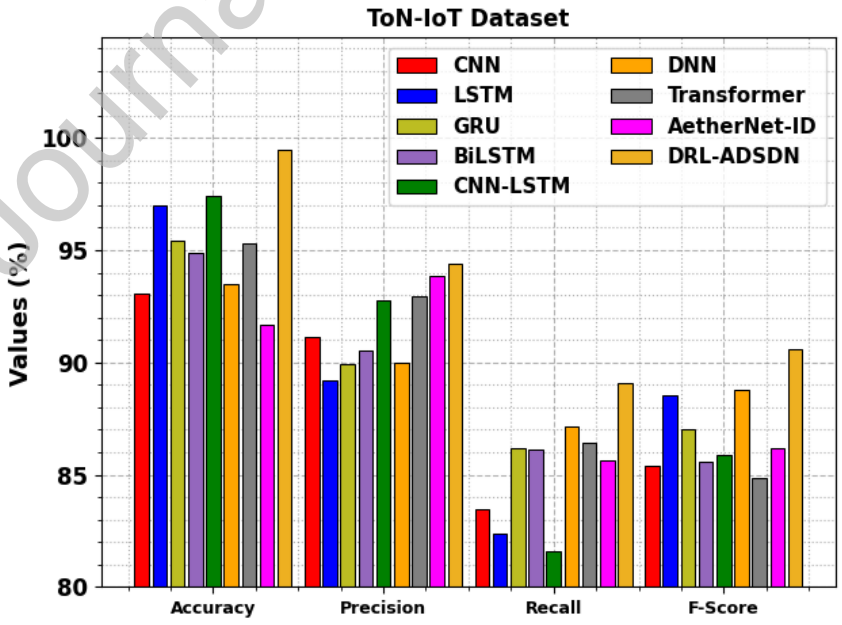


Fig. 15. Comparison assessment of the DRL-ADSDN approach under ToN-IoT dataset

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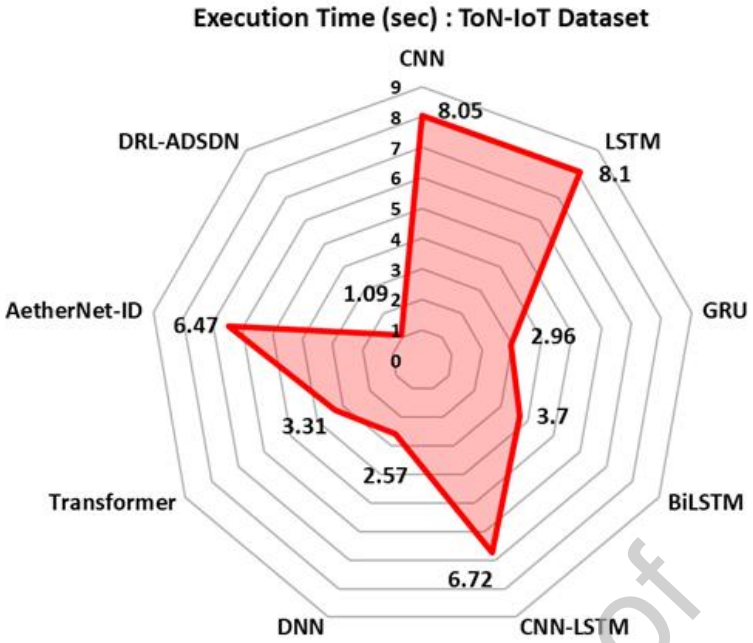


Fig. 16. EXT outcome of DRL-ADSDN approach under ToN-IoT dataset

Table 6 Comparison assessment of the DRL-ADSDN approach under ToN-IoT dataset

Model	$\boldsymbol{Accur_y}$	$\boldsymbol{Preci_n}$	$\boldsymbol{Rec_l}$	$\boldsymbol{F_{Score}}$	**EXT (sec)**
CNN	93.04	91.12	83.47	85.38	8.05
LSTM	97.00	89.17	82.40	88.52	8.10
GRU	95.41	89.95	86.16	87.01	2.96
BiLSTM	94.87	90.55	86.12	85.56	3.70
CNN-LSTM	97.44	92.79	81.60	85.86	6.72
DNN	93.51	89.99	87.12	88.79	2.57
Transformer	95.32	92.97	86.44	84.86	3.31
AetherNet-ID	91.65	93.85	85.66	86.17	6.47
DRL-ADSDN	99.49	94.42	89.09	90.59	1.09

Table 7 specifies the ablation study evaluation of the DRL-ADSDN method under the ToN-IoT dataset. The CRAN+MIC-mRMR model attained an $accur_y$ of 98.84%, $preci_n$ of 93.87%, rec_l of 88.34%, and F_{score} of 89.99%, highlighting the efficiency of the model. By integrating Adam hyperparameter tuning with CRAN and MIC-mRMR feature selection, the performance improved to an $accur_y$ of 99.49%, $preci_n$ of 94.42%, rec_l of 89.09%, and F_{score} of 90.59%, emphasizing that optimizer fine-tuning improves feature learning, classification accuracy, and overall robustness of the model.

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Table 7 Impact of diverse components of the DRL-ADSDN model

Model	$Accur_y$	$Preci_n$	Rec_l	F_{Score}
CRAN+MIC-mRMR (With feature selection without Adam Hyper parameter tuning)	98.84	93.87	88.34	89.99
DRL-ADSDN (CRAN with MIC-mRMR feature selection and Adam Hyper parameter tuning)	99.49	94.42	89.09	90.59

The attack-detection outcomes of the DRL-ADSDN model on the CICDDoS-2019 dataset are shown in Table 8 under the CICDDoS-2019 dataset. The investigational outcome suggests the DRL-ADSDN model is effective in detection. With 70%TRAP, the DRL-ADSDN model attains an average $accur_y$ of 99.46%, $preci_n$ of 95.11%, rec_l of 95.10%, F_{score} of 95.09%, and AUC_{score} of 97.40%. Furthermore, with 30%TESP, the DRL-ADSDN approach attains an average $accur_y$ of 99.49%, $preci_n$ of 95.43%, rec_l of 95.33%, F_{score} of 95.36%, and AUC_{score} of 97.53%.

Table 8 Attack detection of DRL-ADSDN approach under CICDDoS-2019 dataset

Class Labels	$Accur_y$	$Preci_n$	Rec_l	F_{Score}	AUC_{Score}
TRAP (70%)					
Benign	99.51	97.45	93.99	95.69	96.92
DrDoS_DNS	99.40	95.63	93.45	94.52	96.60
DrDoS_LDAP	99.44	95.17	94.31	94.74	97.02
DrDoS_MSSQL	99.40	94.72	94.72	94.72	97.20
DrDoS_NTP	99.52	97.08	94.33	95.69	97.08
DrDoS_NetBIOS	99.37	92.10	96.85	94.41	98.18
DrDoS_SNMP	99.49	95.54	95.54	95.54	97.64
DrDoS_UDP	99.43	93.68	95.88	94.77	97.76
LDAP	99.57	96.87	95.51	96.18	97.66
MSSQL	99.56	95.09	96.76	95.92	98.24
NetBIOS	99.32	94.12	93.85	93.99	96.75
Portmap	99.40	94.94	93.82	94.38	96.77
Syn	99.25	93.06	93.33	93.20	96.47
TFTP	99.59	96.26	96.26	96.26	98.02
UDP	99.59	95.57	97.18	96.37	98.46
UDP-lag	99.41	94.07	95.42	94.74	97.53
UDPLag	99.52	96.81	94.62	95.70	97.22
WebDDoS	99.43	93.75	95.93	94.83	97.78
Average	**99.46**	**95.11**	**95.10**	**95.09**	**97.40**
TESP (30%)					
Benign	99.52	99.19	91.04	94.94	95.50
DrDoS_DNS	99.15	92.00	92.62	92.31	96.07
DrDoS_LDAP	99.52	95.81	96.39	96.10	98.05
DrDoS_MSSQL	99.56	95.71	95.71	95.71	97.74
DrDoS_NTP	99.56	94.70	97.28	95.97	98.48
DrDoS_NetBIOS	99.59	97.30	95.36	96.32	97.60
DrDoS_SNMP	99.63	95.17	97.87	96.50	98.80
DrDoS_UDP	99.33	92.77	96.25	94.48	97.89
LDAP	99.59	97.16	95.14	96.14	97.49

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MSSQL	99.67	96.32	98.12	97.21	98.94
NetBIOS	99.52	96.40	94.37	95.37	97.09
Portmap	99.59	95.71	97.50	96.59	98.61
Syn	99.52	96.71	94.84	95.77	97.32
TFTP	99.44	95.36	94.74	95.05	97.23
UDP	99.44	93.92	95.86	94.88	97.75
UDP-lag	99.41	95.30	94.04	94.67	96.88
UDPLag	99.48	95.24	95.24	95.24	97.48
WebDDoS	99.22	92.99	93.59	93.29	96.58
Average	**99.49**	**95.43**	**95.33**	**95.36**	**97.53**

Fig. 17 investigates the comparative error rate outcomes of the DRL-ADSDN approach with the CICDDoS-2019 dataset. The results showed that the BiLSTM, CNN-LSTM, and DNN techniques achieved the lowest error rates of 8.86, 5.76, and 4.47, respectively. Concurrently, the Transformer and CNN techniques have achieved moderately lower error rates of 4.12 and 3.20, respectively. Moreover, the AetherNet-ID, GRU, and LSTM models have achieved considerable results, with error rates of 2.55, 1.03, and 1.93, respectively. Finally, the DRL-ADSDN method demonstrates a promising solution with the least error rate of 0.51.

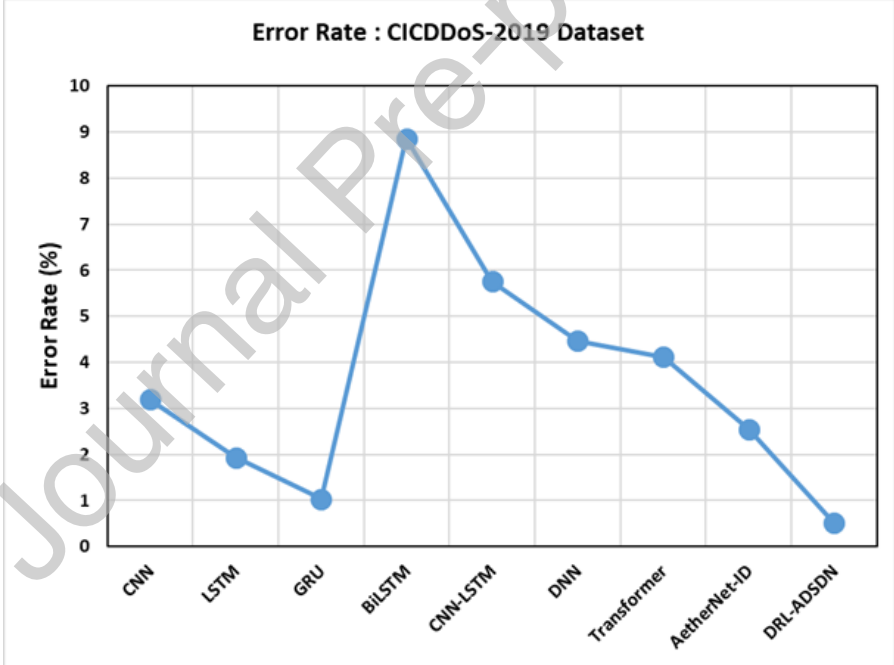


Fig. 17. Error rate of DRL-ADSDN approach under CICDDoS-2019 dataset

Table 9 and Figs. 18-19 exemplify the comparison and EXT evaluation of the DRL-ADSDN method under the CICDDoS-2019 dataset. The results show that the DRL-ADSDN method achieves higher $accur_y$, $preci_n$, rec_l, F_{score}, and EXT of 99.49%, 95.43%, 95.33%, 95.36%, and 1.09s, while conversely, the CNN, LSTM, GRU, BiLSTM, CNN-LSTM, DNN, Transformer, and AetherNet-ID approaches attained lesser values.

Table 9 Comparison analysis of DRL-ADSDN method under CICDDoS-2019 dataset

Model	$Accur_y$	$Preci_n$	Rec_l	F_{Score}	**EXT (sec)**
CNN	93.04	91.12	83.47	85.38	8.05
LSTM	97.00	89.17	82.40	88.52	8.10
GRU	95.41	89.95	86.16	87.01	2.96
BiLSTM	94.87	90.55	86.12	85.56	3.70
CNN-LSTM	97.44	92.79	81.60	85.86	6.72
DNN	93.51	89.99	87.12	88.79	2.57
Transformer	95.32	92.97	86.44	84.86	3.31
AetherNet-ID	91.65	93.85	85.66	86.17	6.47
DRL-ADSDN	99.49	95.43	95.33	95.36	1.09

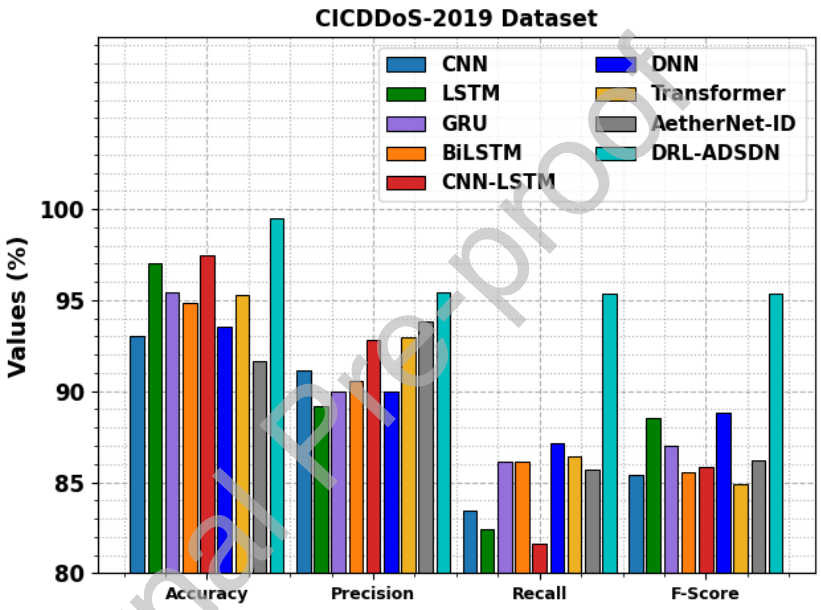


Fig. 18. Comparison analysis of DRL-ADSDN method under CICDDoS-2019 dataset

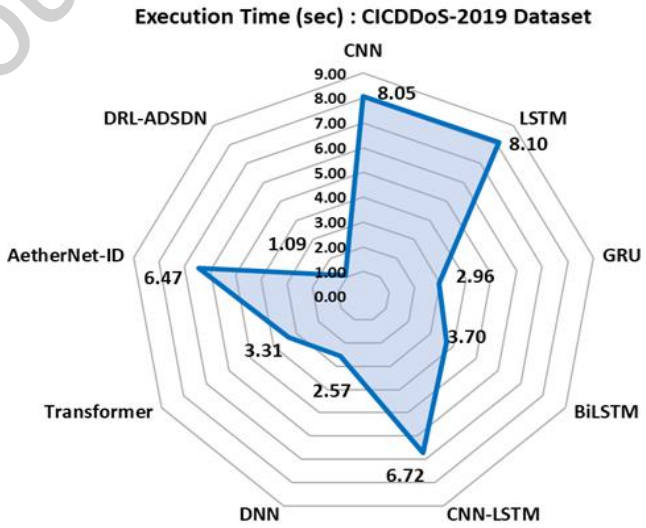


Fig. 19. EXT outcome of DRL-ADSDN approach under CICDDoS-2019 dataset

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The execution-time results give a model-level indication of the computational behaviour in the experiment set-up. The purpose of the MIC-mRMR stage is to reduce the dimensionality data that is fed into the next stage, the CRAN architecture, which carries out convolutional transformations, bidirectional recurrent updates and calculations of attention-weights. This observed execution time is thus the joint effect of feature selection and deep representation learning given the experimental setup of hardware and dataset.

Fig. 20 demonstrates the training (TRAN) and validation (VALD) $accur_y$ of the DRL-ADSDN approach under three datasets over 25 epochs and CICDDoS-2019 dataset. Both curves gradually rise and converge more closely, indicating that the method effectively learns. The VALD $accur_y$ consistently stays slightly higher than the TRAN $accur_y$, suggesting that the model is not overfitting and generalising well to unseen data. The fluctuations in accuracy are expected owing to the task's complexity, but the overall upward trends indicate robust outcomes and method stability.

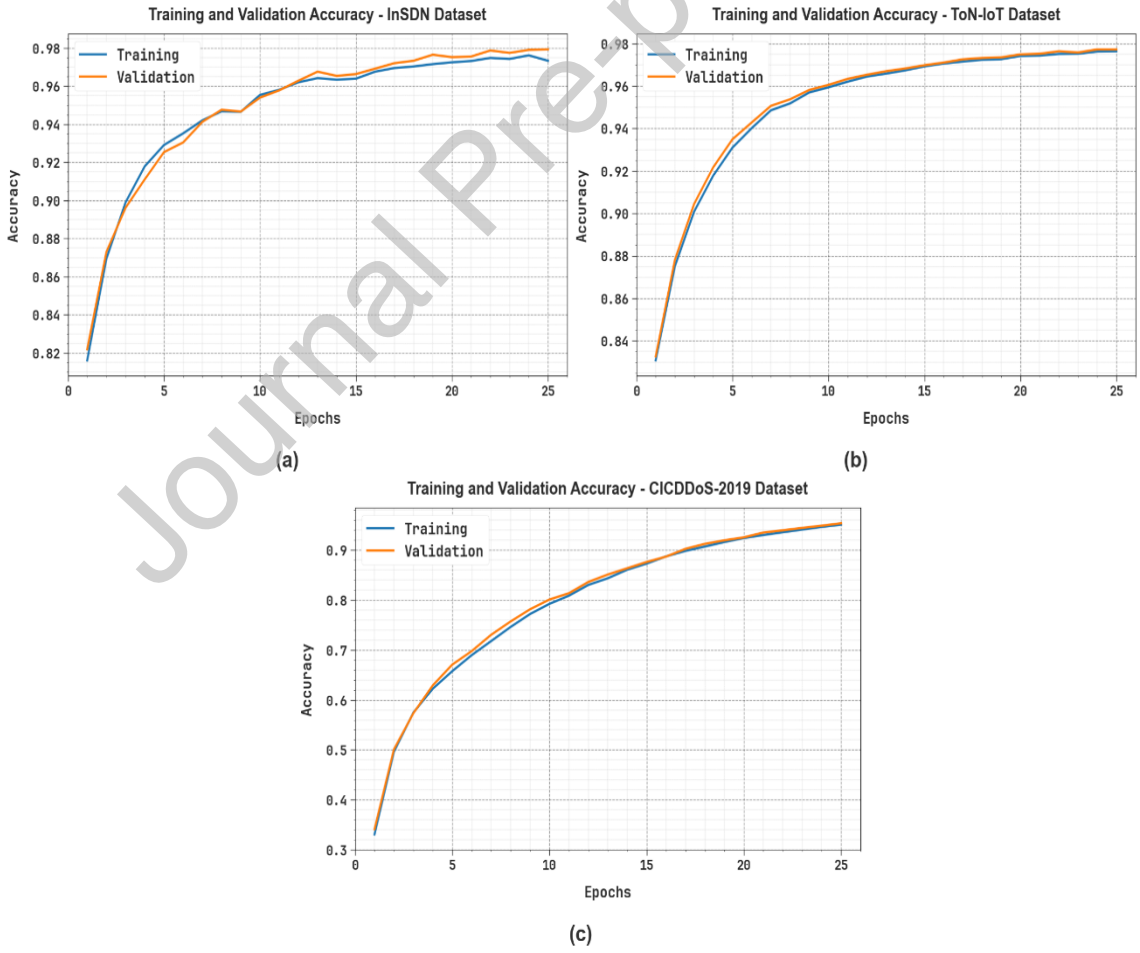


Fig. 20. $Accur_y$ curve of (a) InSDN dataset, (b) ToN-IoT dataset, and (c) CICDDoS-2019 dataset

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Fig. 21 shows the TRAN and VALD losses of the DRL-ADSDN approach across three datasets over 25 epochs under the CICDDoS-2019 dataset. Both curves show a consistent downward trend, indicating that the model effectively reduces error during learning. The VALD loss remains lower than the training loss throughout most epochs, indicating better generalisation and no signs of overfitting. However, some fluctuations were identified to become progressively steady and consistent.

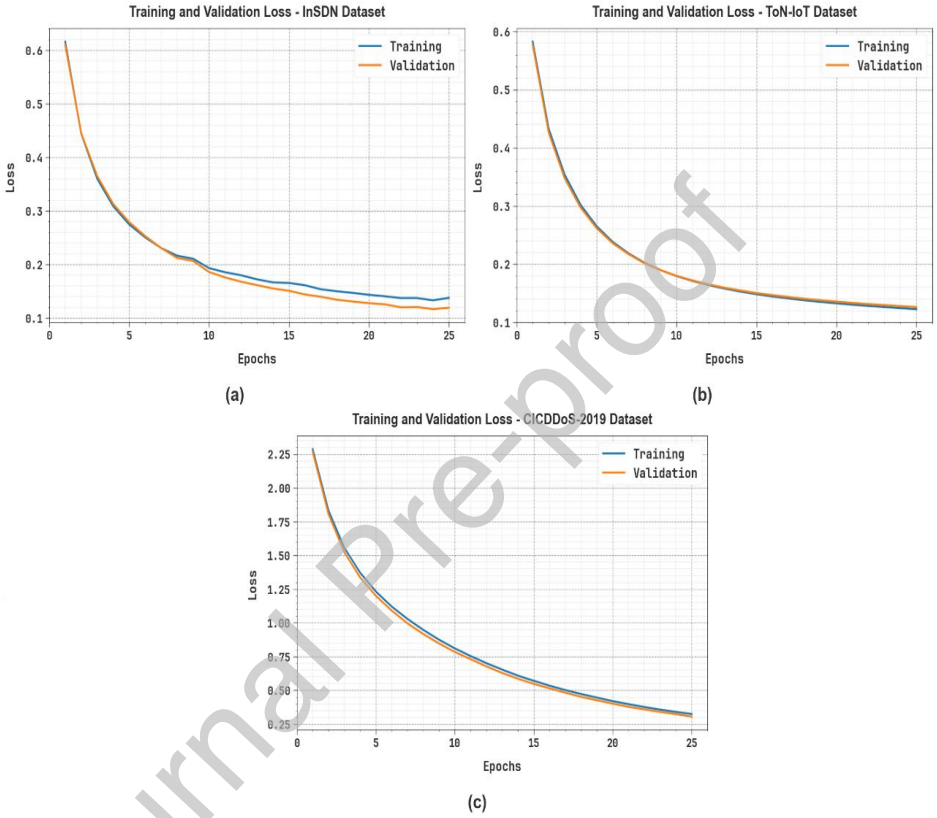


Fig. 21. Loss curve of (a) InSDN dataset, (b) ToN-IoT dataset, and (c) CICDDoS-2019

Table 10 portrays the ablation study evaluation of the DRL-ADSDN approach under the CICDDoS-2019 dataset. The CRAN+MIC-mRMR model with feature selection but without Adam hyperparameter tuning attained an $accur_y$ of 98.83%, $preci_n$ of 94.73%, rec_l of 94.63%, and F_{score} of 94.67%, serving as the baseline performance. The DRL-ADSDN model with CRAN and MIC-mRMR feature selection and Adam hyperparameter tuning attained an $accur_y$ of 99.49%, $preci_n$ of 95.43%, rec_l of 95.33%, and F_{score} of 95.36%, highlighting that optimizer tuning effectively improves model learning, feature utilization, and overall classification performance.

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Table 10 Performance evaluation of CRAN models before and after Adam hyperparameter tuning under the CICDDoS-2019 dataset

Model	$Accur_y$	$Preci_n$	Rec_l	F_{Score}
CRAN+MIC-mRMR (With feature selection without Adam Hyper parameter tuning)	98.83	94.73	94.63	94.67
DRL-ADSDN (CRAN with MIC-mRMR feature selection and Adam Hyper parameter tuning)	99.49	95.43	95.33	95.36

Therefore, the DRL-ADSDN model is effectively employed for accurate and reliable attack detection and classification in SDN.

The comparison also shows that a single hybrid architecture might not be able to deliver uniform detection performance in case of heterogeneous attack categories. CNN based models model local traffic patterns but might not explicitly remove redundant input attributes, while recurrent models model longer temporal relationships but might not have enough ability to model local traffic patterns. While attention mechanisms can highlight important representations, they rely on the quality of the feature representation. The limitations are more apparent when the traffic distribution is not similar in different datasets and attack categories have similar flow characteristics. DRL-ADSDN tackles these complementary problems in turn: (1) MIC-mRMR is applied prior to CRAN, (2) CNN is used for local representation extraction, (3) BiGRU is used for bidirectional temporal modelling and (4) additive attention is used for weighting the representations that are relevant for the attack.

6. Limitations

The present results should be interpreted with some caution due to certain limitations. First, the evaluation is carried out using benchmark datasets and thus will not produce all the characteristics of operational SDN traffic. Second, the current experiments are not repeated run statistical testing & cross validation, and hence there is no quantification of performance variability. Thirdly, the framework has not been assessed specifically with respect to attack types it has not encountered before or the distribution of attacks that have changed. Fourth, computation latency is reported for comparison with computation but is at the model level and not at the end-to-end inference latency in an operational SDN controller. Fifth, results of performance variations across classes, notably the lower performance of MiTM on ToN-IoT, suggests that attack categories are still difficult. Lastly, the present study does not explicitly

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consider the use of controller resources, OF control-plane telemetry, multi-controller synchronisation or high throughput traffic conditions.

7. Conclusion

This manuscript presented the DRL-ADSDN framework to identify and classify attacks in SDN environments. The framework combines the pre-processing, dimensionality reduction using MIC-mRMR, spatial-temporal representation learning using CRAN, and RAdam optimization. The evaluation done with InSDN, ToN-IoT and CICDDoS-2019 shows the same detection ability where the minimum error rate of each one is 0.69, 0.51 and 0.51 respectively. The analysis of the execution times also shows the computational efficiency of the evaluated models compared to the baseline models. InSDN is the main validation tool for SDN, with complementary cross-dataset validation carried out by ToN-IoT and CICDDoS-2019. The proposed framework is aimed at being a traffic level intrusion detection layer, and it does not need to be implemented on any specific SDN controller or switch. However, other issues in the current experimental setup were not addressed, such as live controller deployment, OpenFlow control-plane telemetry, controller placement, multi-controller synchronization, controller resource usage, and high-throughput evaluation. These factors provide valuable directions for future validations. The proposed framework will be extended toward realistic SDN deployments via experimentation with the integrated controller and emulated SDN environments in the future. In the next stage, centralized and distributed multi-controller architectures, controller placement, inter-controller synchronization and controller resource utilization will be explored. Additional control-plane capabilities derived from OpenFlow can be added, in addition to the existing flow-level representation. Further testing will explore the latency for inferences, throughput and scalability as traffic levels increase. In addition, online adaptation mechanisms will be analysed for adapting to changing traffic distributions and attacks that have not been seen before. These extensions will allow for a more complete evaluation of the proposed framework in operational SDN.

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Data Availability Statement: The data that support the findings of this study are openly available in the Kaggle repository at https://www.kaggle.com/datasets/badcodebuilder/insdn-dataset, https://www.kaggle.com/datasets/arnobbhowmik/ton-iot-network-dataset/data, https://www.kaggle.com/datasets/tarundhamor/cicids-2019-dataset/data, and reference numbers [33-35].

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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